IRRATIONAL POWERS ON ONE PAGE

# $\pi^{\mathrm{e}}$ or $\mathrm{e}^{\pi}$, and the 23 between them

An irrational power is defined through $\ln$, and the same logarithm settles which of two famous powers is larger.

THE RULE

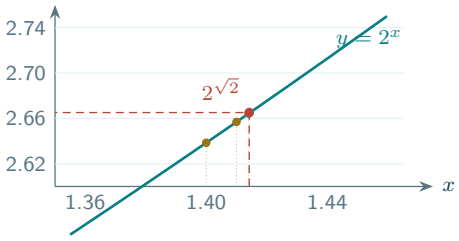
## An irrational power is a limit, or a logarithm

A rational power is a root of a whole power, $2^{7/5} = \sqrt[5]{2^7}$. An irrational exponent such as $\sqrt{2}$ is approached by its decimal truncations, and because $2^x$ increases steadily the powers settle on one value,

$$2^{1.4} = 2.639, \quad 2^{1.41} = 2.657, \quad 2^{1.414} = 2.6647, \quad \ldots \to 2^{\sqrt{2}} \approx 2.66514.$$

The same number comes from a single formula for every real $y$ and base $x > 0$, which turns any comparison of powers into one of logarithms:

$$x^y \quad = \quad \mathrm{e}^{\,y \ln x}$$




Truncations of $\sqrt{2}$ climb the graph of $2^x$ towards $2^{\sqrt{2}}$.

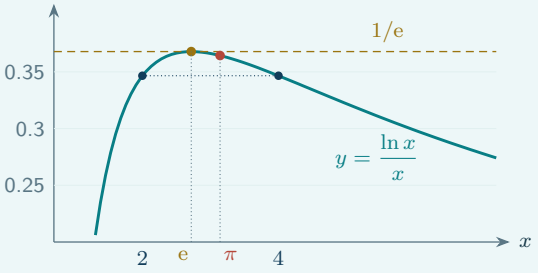
WHICH IS BIGGER?

## Compare $\dfrac{\ln x}{x}$, which peaks at $x = \mathrm{e}$

Taking $\ln$ of both numbers gives $\ln(\pi^{\mathrm{e}}) = \mathrm{e}\ln\pi$ and $\ln(\mathrm{e}^{\pi}) = \pi$, and dividing both by $\mathrm{e}\pi$ reduces the question to comparing $\frac{\ln\pi}{\pi}$ with $\frac{\ln\mathrm{e}}{\mathrm{e}}$. The function $f(x) = \frac{\ln x}{x}$ has $f'(x) = \frac{1-\ln x}{x^2}$, which vanishes only at $x = \mathrm{e}$, so its single maximum is $f(\mathrm{e}) = \frac{1}{\mathrm{e}}$ and every other $x > 0$ has $f(x) < \frac{1}{\mathrm{e}}$, that is $x^{\mathrm{e}} < \mathrm{e}^x$. Putting $x = \pi$,

$$\pi^{\mathrm{e}} \quad \approx \quad 22.459 \quad < \quad \mathrm{e}^{\pi} \quad \approx \quad 23.141$$

The margin is thin because $\pi$ sits just past the peak, where $f(\pi) = 0.3644$ against $f(\mathrm{e}) = 0.3679$. The same graph explains the only whole-number coincidence of this kind, $2^4 = 4^2$, since $f(2) = f(4)$.




The peak at $\mathrm{e}$ beats every other value, $\pi$ only narrowly; 2 and 4 share a height.

GOOD GUESSES

## Both by hand, from $\mathrm{e}^3 \approx 20.09$

Splitting each exponent at 3 leaves a small $t$ for $\mathrm{e}^t \approx 1 + t + \frac{t^2}{2}$:

$$\mathrm{e}^{\pi} = \mathrm{e}^3\,\mathrm{e}^{0.1416} \approx 20.09 \times 1.152 \approx 23.14,$$

and with $\ln\pi \approx \ln 3 + 0.046 \approx 1.145$, so that $\mathrm{e}\ln\pi \approx 3.112$,

$$\pi^{\mathrm{e}} = \mathrm{e}^3\,\mathrm{e}^{0.112} \approx 20.09 \times 1.118 \approx 22.46.$$

WHAT IS KNOWN

## One is proved transcendental, one is open

Since $\mathrm{e}^{\pi} = (-1)^{-\mathrm{i}}$, an algebraic number raised to a non-rational algebraic power, the Gelfond–Schneider theorem makes Gelfond's constant $\mathrm{e}^{\pi}$ transcendental. Nobody has yet proved that $\pi^{\mathrm{e}}$ is even irrational. A near miss worth knowing is $\mathrm{e}^{\pi} - \pi = 19.99910\ldots$, which has no known explanation.

THE ONE INTEGER IN THE GAP

## $22.459 < 23 < 23.141$, and 23 is where cyclotomic factorisation first fails

The interval between the two powers is only $0.68$ wide and holds exactly one whole number, 23. In 1847 Lamé announced a proof of Fermat's Last Theorem that factorised $x^p + y^p = \prod_k (x + \zeta^k y)$ in the cyclotomic integers $\mathbb{Z}[\zeta_p]$, with $\zeta_p = \mathrm{e}^{2\pi\mathrm{i}/p}$, and silently assumed that factorisation there is unique. Kummer showed the assumption holds for every prime $p \le 19$ and fails first at $p = 23$, where the class number of $\mathbb{Q}(\zeta_{23})$ is 3; he rescued unique factorisation by inventing ideal numbers, the ancestors of modern ideals.

$$h\big(\mathbb{Q}(\zeta_p)\big) \quad = \quad 1 \text{ for } p \quad = \quad 2, 3, 5, 7, 11, 13, 17, 19, \qquad h\big(\mathbb{Q}(\zeta_{23})\big) \quad = \quad 3$$

The class number is the one from the Heegner sheet, where $h = 1$ for $\mathbb{Q}(\sqrt{-163})$ gave forty primes in a row; that 23 also falls between $\pi^{\mathrm{e}}$ and $\mathrm{e}^{\pi}$ is an accident of size.

**Conventions.** For a positive base, $x^y = \mathrm{e}^{y\ln x}$ is the unique real value and agrees with roots and integer powers whenever $y$ is rational; for a negative or complex base the logarithm has many branches and the value depends on the branch chosen. The class number $h$ measures how far a ring of integers is from unique factorisation, with $h = 1$ meaning it has it.

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